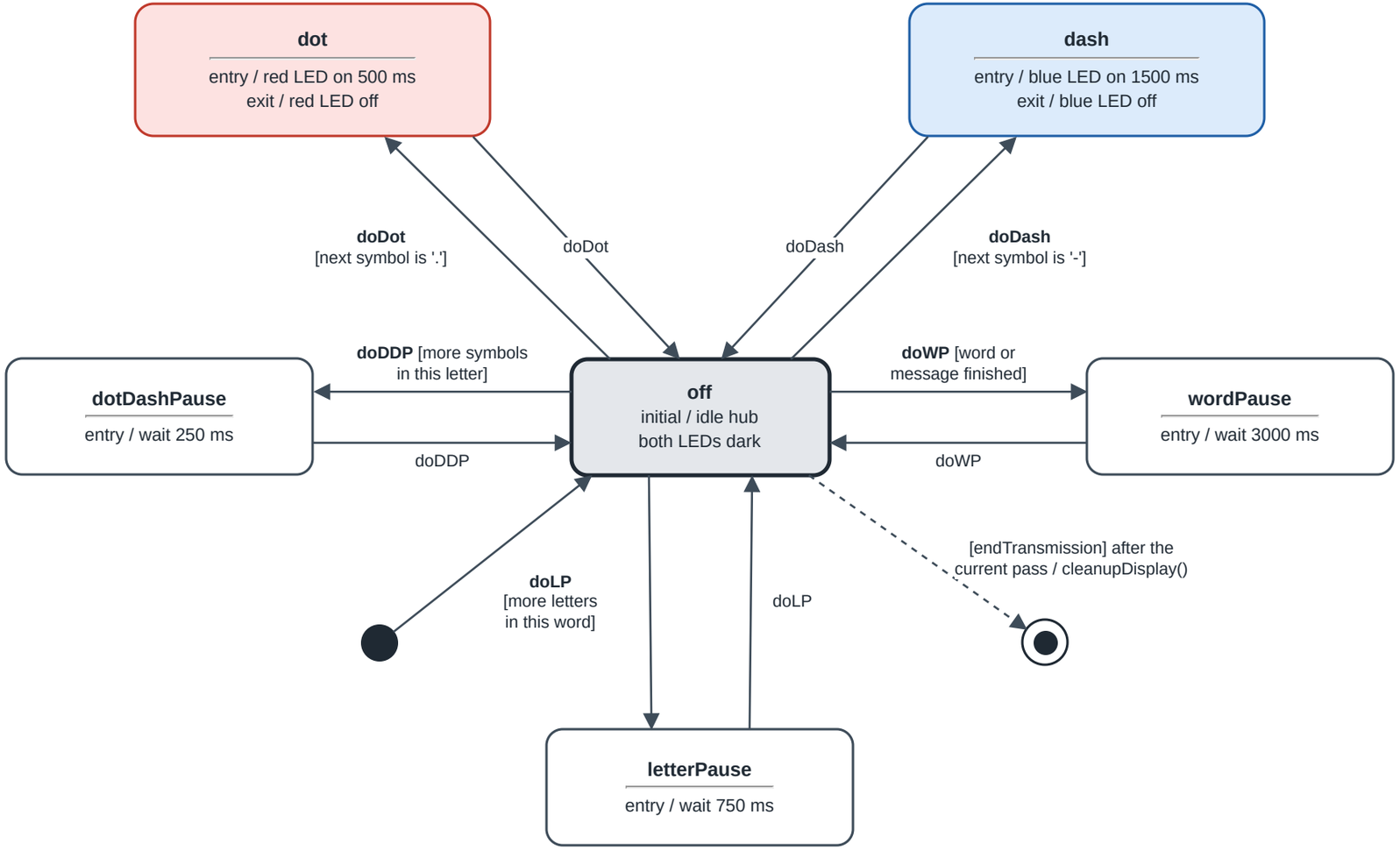
Milestone Three: Morse Code State Machine

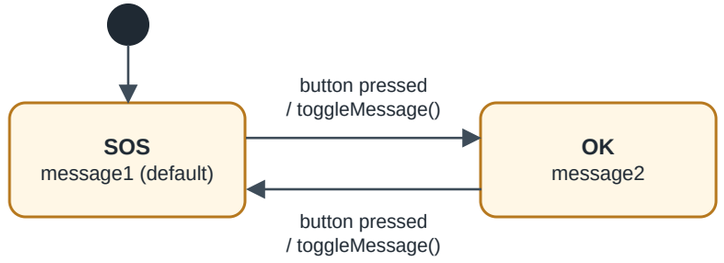
Acram Kiyingi · CS-350 Emerging Systems Architecture · Milestone3.py on Raspberry Pi 4 · red LED (GPIO 18) = dot, blue LED (GPIO 23) = dash, button on GPIO 24

CWMachine: one state per Morse element, all routed through *off*



Every event is a two-way pair (off → X, then X → off), so each element starts and ends in *off* with both LEDs dark. A pass of **SOS** visits 9 symbol states, 6 dot/dash pauses, 2 letter pauses and 1 word pause: 13.5 s.

Message selector: green button on GPIO 24



The transmit thread reads *activeMessage* only at the start of each pass, so a press never interrupts the message being sent. The new message starts on the next pass.

Timing and runtime

Dot (red, GPIO 18)	500 ms
Dash (blue, GPIO 23)	1500 ms
Between dots/dashes	250 ms
Between letters	750 ms
Between words / before a repeat	3000 ms

16x2 LCD: shows `Sending:` / `<activeMessage>` at the start of every pass.

Threads: the transmit thread drives CWMachine; the gpiozero callback runs toggleMessage(); the main thread idles until Ctrl-C sets endTransmission.